

Bayesian Optimisation with Replication Selection for Stochastic Simulation Optimisation

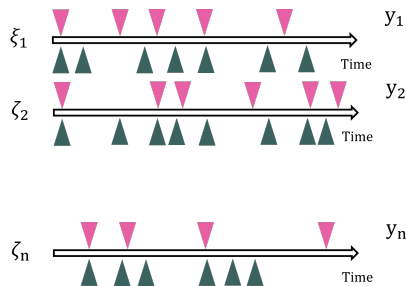
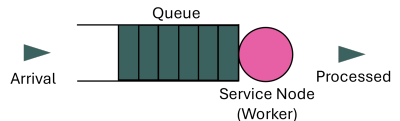
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Stochastic Simulation

Stochastic simulation is used in the modelling and optimisation for a number of complicated problems:

- ▶ Hospital treatment pathways
- ▶ Inventory management
- ▶ Supply chain optimisation
- ▶ And many more....



▼ Service Event

▲ Arrival Event

y_i Performance Metric

$$\bar{y} = \frac{1}{n} \sum_{i=0}^n y_i$$

Stochastic Simulation

Evaluations of a simulation at a given decision x are noisy samples of the true performance $f(x)$.

$$y(x) = f(x) + \varepsilon, \quad \varepsilon \sim N(0, \sigma_\varepsilon^2)$$

Therefore we usually take n replications at each decision for a better estimate of $f(x)$

$$\bar{f}(x) \sim N\left(f(x), \frac{\sigma_\varepsilon^2}{n}\right)$$

Optimisation

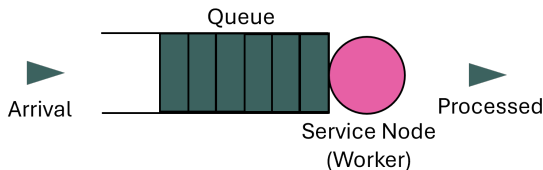
We view stochastic simulation optimisation within the context of a black-box optimisation problem:

$$f^*(x) = \max_{x \in X} \mathbb{E}[y(x)]$$

Heteroscedastic Noise

In the general case, the noise variance $\sigma_\varepsilon^2 \rightarrow \sigma_\varepsilon^2(x)$ is not constant across the decision space. This is known as *heteroscedastic* noise.

Consider the MM1 for example with arrival rate λ and service rate μ .



- ▶ As $\lambda \rightarrow \mu$, the system is unstable and the noise variance is very high.
- ▶ When $\lambda \ll \mu$, the system is stable and the noise variance is low.

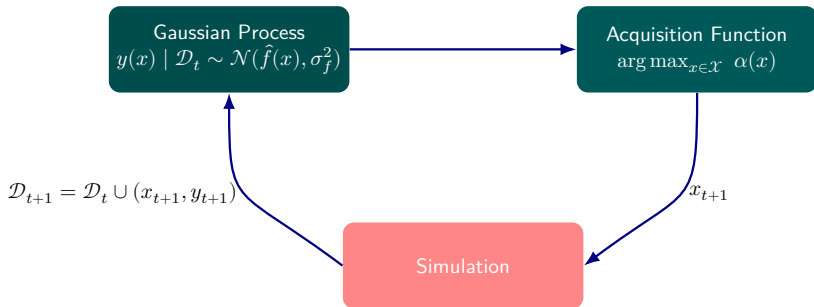
Bayesian Optimisation (BO)

Bayesian Optimisation (BO) performs global optimisation using informed decisions to explore $\mathcal{X}$.

BO is built upon the foundations of its two main components:

- ▶ **Surrogate Model** - The algorithm's present knowledge about the unknown objective; used to predict **mean** and **uncertainty** values of unobserved points. Usually a Gaussian Process
- ▶ **Acquisition Function (AF)** - How BO evaluates the utility of any $x \in \mathcal{X}$ in its effort to find f^*

Bayesian Optimisation (BO)



BO with replication selection

Aim: Tailor a BO algorithm toward stochastic simulation optimisation, by considering heteroscedastic noise in the GP and including, as an additional decision, the number of replications at which to sample a point.

Heteroscedastic Gaussian Process

Observations from the simulation are modelled with a Gaussian Process (GP)(Lázaro-Gredilla and Titsias 2011):

$$y(x) \sim \mathcal{GP}(f(x), \sigma_{\varepsilon}^2(x))$$

$$f(x) \sim \mathcal{GP}(\mu_f, k_f(x, x')), \quad g(x) \sim \mathcal{GP}(\mu_g, k_g(x, x'))$$

$$\sigma_{\varepsilon}^2(x) = \phi(g(x))$$

$\mu_{(\cdot)}$ is the prior mean function; it's usually constant

$k_{(\cdot)}(x, x)$ is the kernel function

$\phi(x)$ is a link function to enforce positive noise

Replication Aware Acquisition Function

Our AF is based upon (Moss et al. 2021), which is a lower bound on the max-entropy search.

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$$\alpha_{\text{IG}}(x, n) = -\frac{1}{2l}c(n) \sum_{i=1}^l \log \left(1 - \frac{n\sigma_f^2(x)}{\sigma_\epsilon^2(x) + n\sigma_f^2(x)} Z(x, f_i^*) \left[\gamma(x, f_i^*) + Z(x, f_i^*) \right] \right)$$

$$x_{t+1}, n_{t+1} = \underset{x \in \mathcal{X} \quad n \in \mathcal{N}}{\operatorname{argmax}} \alpha(x, n)$$

Inverse Cost

$$\alpha_{\text{IG}}(x, n) = -\frac{1}{2l} c(n) \sum_{i=1}^l \log \left(1 - \frac{n \sigma_f^2(x)}{\sigma_\varepsilon^2(x) + n \sigma_f^2(x)} Z(x, f_i^*) \left[\gamma(x, f_i^*) + Z(x, f_i^*) \right] \right)$$

$$c(n) = \frac{1}{\beta_0 + \beta_1 n}$$

β_0 is the base cost of running the simulation

β_1 is the per replication cost

Noise Penalty

$$\alpha_{\text{IG}}(x, n) = -\frac{1}{2l}c(n) \sum_{i=1}^l \log \left(1 - \frac{n\sigma_f^2(x)}{\hat{\sigma}_\varepsilon^2(x) + n\sigma_f^2(x)} Z(x, f_i^*) \left[\gamma(x, f_i^*) + Z(x, f_i^*) \right] \right)$$

$|\hat{\sigma}_\varepsilon^2(x)$ is the predicted *aleatoric uncertainty* $|\sigma_f^2(x)$ is the GP's *epistemic uncertainty*

f^* Improvement term

$$\alpha_{\text{IG}}(x, n) = -\frac{1}{2l}c(n) \sum_{i=1}^l \log \left(1 - \frac{n\sigma_f^2(x)}{\hat{\sigma}_\varepsilon^2(x) + n\sigma_f^2(x)} Z(x, f_i^*) \left[\gamma(x, f_i^*) + Z(x, f_i^*) \right] \right)$$

$$\gamma(x, f_i^*) = \frac{f_i^* - \hat{f}(x)}{\sigma_f^2(x)}$$

$$Z(x, f_i^*) = \frac{\Phi(\gamma(x, f_i^*))}{\Psi(\gamma(x, f_i^*))}$$

$\hat{f}(x)$ is the predicted mean at x

Φ is the standard normal pdf

Ψ is the standard normal cdf

Approximation

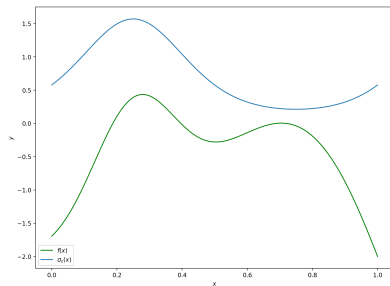
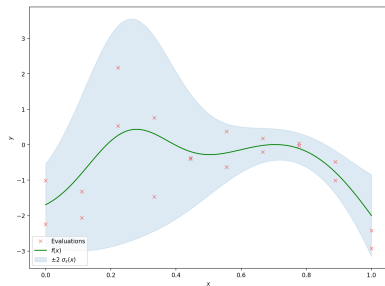
$$\alpha_{\text{IG}}(x, n) = -\frac{1}{2l}c(n)\sum_{i=1}^l \log \left(1 - \frac{n\sigma_f^2(x)}{\hat{\sigma}_\varepsilon^2(x) + n\sigma_f^2(x)} Z(x, f_i^*) \left[\gamma(x, f_i^*) + Z(x, f_i^*) \right] \right)$$

$$f^* \sim \text{Gumbel}(a, b)$$

Where a and b are parameters which best fit GP predictions at a subset of locations within the dataset $\mathcal{D}$

Test Problem

Test Problem w/ evaluations for $m = 0$



Problem also used in (Binois and Gramacy 2021)

Experiment parameters

Table 1: Experiment parameters

Parameter	Value
k	10
n_{init}	2
M	50

Table 2: Cost function parameters

Parameter	Value
β_0	1
β_1	0.5

- ▶ $\mathcal{D}_{\text{init}}$ is composed of $B_{\text{init}} = k \times n_{\text{init}}$, at k evenly space unique locations, each with n_{init} replicates.
- ▶ The results are calculated from running BO over M unique

Comparison

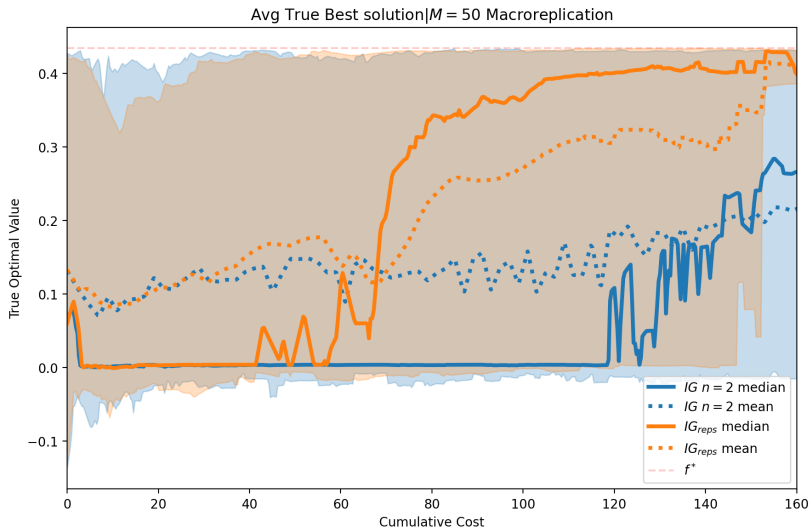
Name	GP	AF	Iterations
IG_{reps}	HGP	$\alpha(x, n)$	$T = 25$
IG	HGP	$\alpha(x, n = 2)$	$T = 80$

Performance

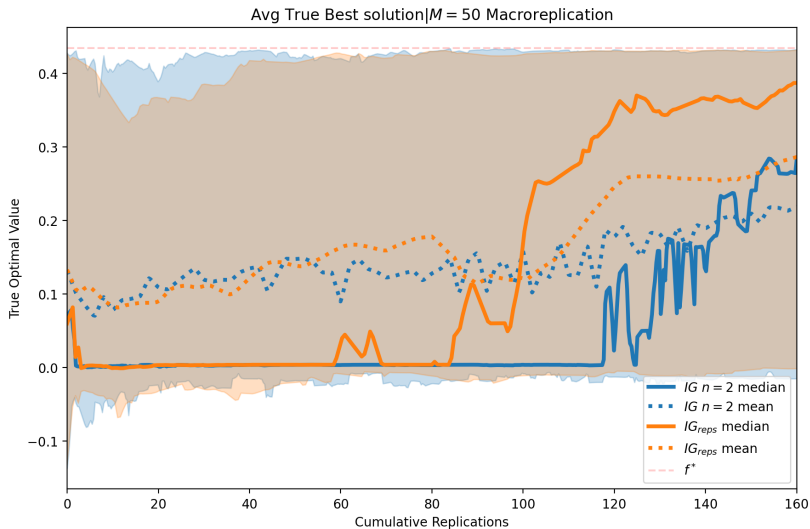
Efficiency A method's proximity to the true optimal solution given the total cumulative cost of the simulation queries it spent to reach that point.

True optimal solution The noiseless output of the true test function $f(x_t^*)$ evaluated at the predicted maximiser $x_t^* = \operatorname{argmax}_{x \in \mathcal{X}} \hat{f}(x)$

Results (Cumulative Cost)

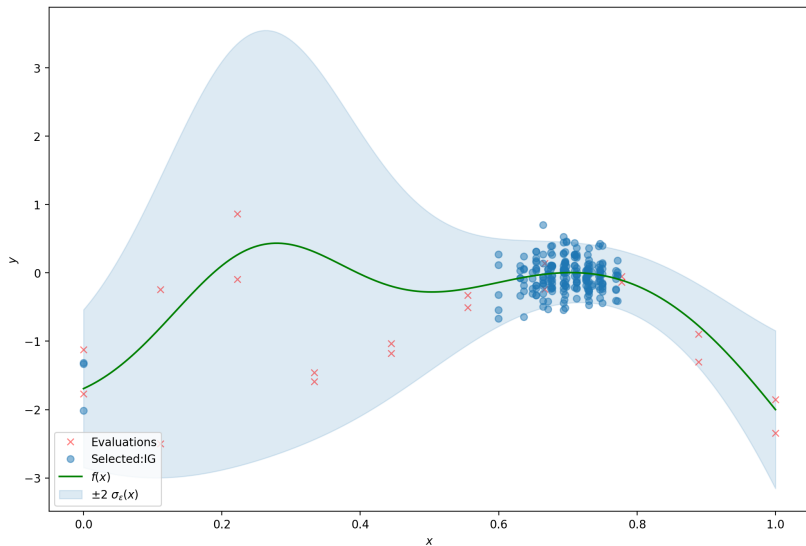


Results (Cumulative Replications)



Local Optima

Candidate selection w/ evaluations for $m = 10$

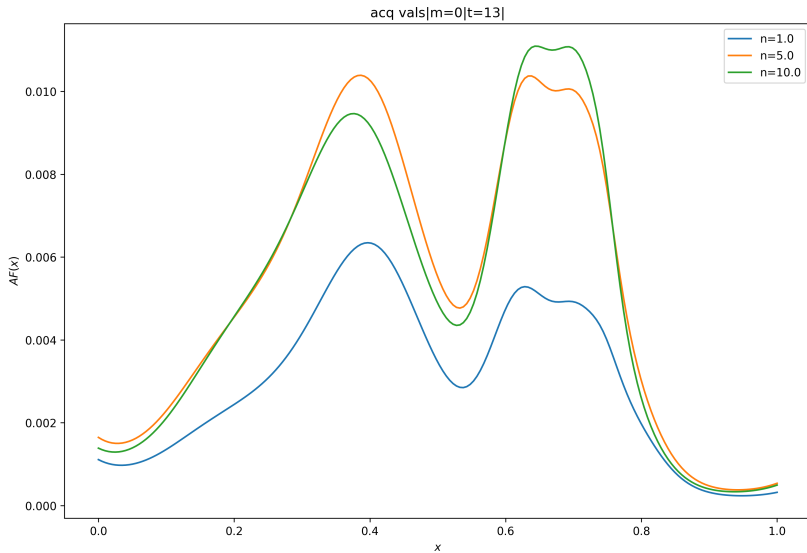


Why?

$$\frac{n\sigma_f^2(x)}{\sigma_\epsilon^2(x) + n\sigma_f^2(x)}$$

- ▶ Preference to explore areas with high $\sigma_f^2(x)$ and low $\hat{\sigma}_\epsilon^2(x)$
- ▶ Increasing n improves the value of high $\hat{\sigma}_\epsilon^2(x)$ areas by inflating $\sigma_f^2(x)$.

Over Exploitation



Fixes?

- ▶ Don't jointly optimise over both x and n . Optimise x first, then make the replication decision
- ▶ Change how the penalty term is parameterised to reduce the gap between aleatoric and epistemic uncertainty

Questions?

References

- Binois, Mickaël, and Robert B. Gramacy. 2021. “**hetGP** : Heteroskedastic Gaussian Process Modeling and Sequential Design in *R*.” *J. Stat. Soft.* 98 (13). <https://doi.org/10.18637/jss.v098.i13>.
- Lázaro-Gredilla, Miguel, and Michalis K. Titsias. 2011. “Variational Heteroscedastic Gaussian Process Regression.” In *International Conference on Machine Learning*. <https://api.semanticscholar.org/CorpusID:8740432>.
- Moss, Henry B., David S. Leslie, Javier Gonzalez, and Paul Rayson. 2021. “GIBBON: General-purpose Information-Based Bayesian Optimisation.” October 26, 2021. <http://arxiv.org/abs/2102.03324>.